Lecture 28

The index on the sphere

Objectives. *Lecture 22 associated an integer with the flux through a sphere. We now use that integer to count quantum zero modes of a Dirac operator. The spin connection changes the line bundle in which one chiral component lives. On the round sphere this turns the count into a polynomial calculation. The cited index theorem explains why the difference of the two chiral counts is topological.*

By the end of this lecture you will be able to:

- Specify the two chiral components and the sign of their index.
- Count the holomorphic sections of $O(m)$ for every $m \in \mathbb{Z}$.
- Compute both monopole zero-mode counts using the stated cited identifications.
- Distinguish the Dirac count from the spinless Landau-level count and state the anomaly convention.

28.1 Spinors and the canonical line on the sphere

Use the unit round sphere with its outward orientation. Identify it with $\mathbb{C}\mathrm{P}^1$ using Lecture 20's complex coordinates z and $w = 1/z$. The complex orientation agrees with this outward orientation. A *spinor field* is a section of a spinor bundle, the geometric bundle on which the Dirac equation acts. Its construction is a cited input here (Lawson–Michelsohn, Chapter II and Appendix D). On an oriented surface it has two complex line components S^+ and S^-. Their labels specify *chirality*, a grading with eigenvalues +1 and −1. Chirality here is a spinor grading, distinct from even and odd differential-form degree.

The *canonical bundle* K_{S^2} is the line of holomorphic cotangent vectors. Its local frames are $\mathrm{d}z$ and $\mathrm{d}w$. Since $\mathrm{d}w = -z^{-2}\mathrm{d}z$, absorbing the constant minus sign into one frame gives $K_{S^2} \cong O(-2)$ in Lecture 22's frame convention. We choose the positive spinor line to be its square root:

$$S^+ = K_{S^2}^{1/2} \cong O(-1), \qquad (K_{S^2}^{1/2})^{\otimes 2} \cong K_{S^2}.$$

This square-root choice is part of the cited spinor identification. We do not construct spin structures in this lecture.

Recall that $L_n = O(n)$ is the monopole line of Chern number $n \in \mathbb{Z}$. Twisting means taking the tensor product with this charged line. Put $E_n = S^+ \otimes L_n \cong O(n-1)$. The full twisted spinor bundle has the cited description

$$S \otimes L_n \cong \left(\Lambda^{0,0} \oplus \Lambda^{0,1}\right) \otimes E_n.$$

Here a $(0,0)$-form is a function and a $(0,1)$-form is locally a multiple of $\mathrm{d}\bar{z}$. The positive part is E_n. The negative part is $\Lambda^{0,1} \otimes E_n$. The figure records the twist that will change the polynomial count.

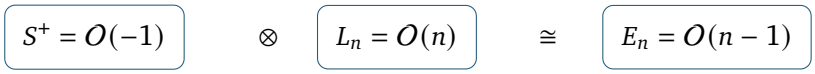


Figure 28.1: The positive spinor line has degree -1. Tensoring with the monopole line of degree n gives degree $n-1$.

These identifications are stated in Lawson–Michelsohn, Appendix D, equation (D.16), p. 396. Walpuski, Propositions 70.3 and 70.6, pp. 76–78, give the form decomposition and square-root construction explicitly.

28.2 The Dirac operator and its index

Write $\Omega^{0,k}(E_n)$ for the smooth E_n-valued $(0,k)$-forms. The *Dolbeault operator* $\bar\partial_A$ is the $(0,1)$ part of the covariant derivative. In a holomorphic frame it sends a coefficient f to $(\partial_{\bar z} f)\mathrm{d}\bar z$, as in Lecture 22. Its adjoint $\bar\partial_A^*$ is defined by the Hermitian L^2 inner products from the bundle metric and sphere area. The first slot of these inner products is antilinear. There are no $(0,2)$-forms on a complex curve.

Proposition 28.1 (The sphere Dirac operator · *sketched*)**.** *With the above chirality, the twisted Dirac operator is*

$$D = \sqrt{2}(\bar\partial_A + \bar\partial_A^*) = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}, \qquad D^+ = \sqrt{2}\bar\partial_A, \qquad D^- = (D^+)^* = \sqrt{2}\bar\partial_A^*.$$

The block D^+ maps $\Omega^{0,0}(E_n)$ to $\Omega^{0,1}(E_n)$. The other block maps in the reverse direction.

Explanation. The cited spinor identification turns Clifford multiplication into exterior multiplication by an antiholomorphic covector and its adjoint contraction. Contracting the spin connection and monopole connection with it gives the displayed operator. This identification, including its $\sqrt{2}$ normalization, is Walpuski's Proposition 70.3. For a flat metric $\mathrm{d}x^2 + \mathrm{d}y^2$, the unit antiholomorphic coframe is $\mathrm{d}\bar z/\sqrt{2}$. Thus the positive block has coefficient $2\partial_{\bar z}$ and the negative block has coefficient $-2\partial_z$. Their product is $-4\partial_z\partial_{\bar z} = -\partial_x^2 - \partial_y^2$, the nonnegative scalar Laplacian of Lecture 15. To compare with the familiar flat Dirac equation, let σ_x and σ_y be the Pauli matrices. The corresponding self-adjoint spatial operator is $-\mathrm{i}(\sigma_x\partial_x + \sigma_y\partial_y)$. Multiplying its negative component by i gives the displayed chiral blocks. This verifies the local normalization and recalls the flat operator, not the global spinor construction. ∎

A *zero mode* is a smooth spinor annihilated by D. We use its self-adjoint realization on the closed sphere with the first-order Sobolev domain defined next. This is supplied by cited elliptic spectral theory (Lawson–Michelsohn, Theorem 5.8, pp. 196–197), with the domain identified by the elliptic estimate in Theorem 5.2. The first-order Sobolev domain consists of square-integrable sections whose first derivatives, understood by integration by parts, are also square-integrable in local charts. Its chiral block is *Fredholm*: between this domain and L^2 it has closed range and finite-dimensional kernel and cokernel. The cokernel is the target modulo the range, and is identified with $\ker D^-$ by the adjoint. These analytic facts follow from cited elliptic theory (Lawson–Michelsohn, Chapter III, Theorem 5.2, pp. 193–194). The *index* is the signed difference

$$\operatorname{ind} D^+ = \dim\ker D^+ - \dim\ker D^-.$$

Both dimensions are over $\mathbb{C}$. An index of zero alone does not say that either kernel vanishes.

Along a norm-continuous path of Fredholm chiral blocks from the fixed Sobolev domain to L^2, the index is constant by the cited Fredholm stability theorem (Lawson–Michelsohn, Proposition 7.1 and Corollary 7.4, pp. 201–202). Each kernel dimension can change, with equal changes in the two counts. The finite-dimensional model $T_t\colon \mathbb{C}^2 \to \mathbb{C}$, given by $T_t(a,b) = ta$, illustrates this: for $t \neq 0$ its kernel and adjoint kernel have dimensions 1 and 0. At $t = 0$ they have dimensions 2 and 1. Their difference stays one. This is a model for index stability, not a claim that these jumps occur within the special sphere family below. Perturbations that destroy the chiral block structure are outside this statement.

The square D^2 preserves chirality. Its positive-energy states pair across the two components: if $D^-D^+u = \epsilon u$ with $\epsilon > 0$, then $D^+u/\sqrt{\epsilon}$ is a normalized state of D^+D^- with the same eigenvalue. Its norm is one because $\|D^+u\|^2 = \langle u, D^-D^+u\rangle = \epsilon$ for $\|u\| = 1$. The reverse map is $D^-/\sqrt{\epsilon}$. The figure shows why a difference can remain only at zero.

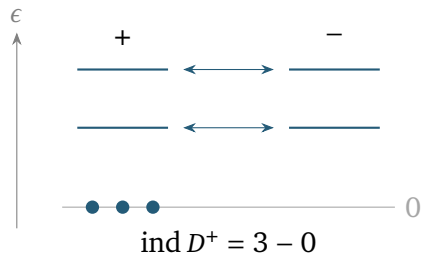


Figure 28.2: Positive squared-Dirac eigenvalues pair between chiralities. The schematic zero eigenspace has three positive modes and no negative mode, so its index is three.

28.3 Polynomial sections and monopole zero modes

For a holomorphic line E, define $H^0(E) = \ker \bar\partial_E$, the space of its global holomorphic sections. Define its Dolbeault cohomology by

$$H^1(E) = \Omega^{0,1}(E)/\bar\partial_E \Omega^{0,0}(E).$$

Every $(0,1)$-form is Dolbeault-closed in complex dimension one, explaining this quotient. The cited Dolbeault Hodge theorem identifies each class with a unique harmonic representative. In this dimension that representative is annihilated by $\bar\partial_E^*$. Consequently the cited spinor and Hodge inputs identify $\ker D^+ \cong H^0(E_n)$ and $\ker D^- \cong H^1(E_n)$ (Walpuski, Proposition 70.3, p. 77).

We now count H^0 directly, including the global condition at infinity. Let $m \in \mathbb{Z}$ and recall the frames $e_S = e_N z^m$ for $O(m)$. A section with coefficients f_N and f_S satisfies

$$f_S(w) = w^m f_N(1/w).$$

Both coefficients must be holomorphic at their own chart origin. The figure displays the monomial test for this condition.

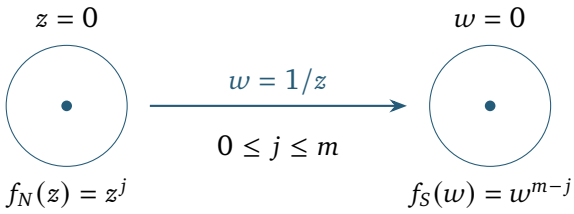


Figure 28.3: A northern monomial z^j becomes w^{m-j} in the southern frame. Holomorphicity at both poles requires $0 \le j \le m$.

Proposition 28.2 (Sections of a degree-m line · *proved given elementary Laurent uniqueness*)**.** *For $m \ge 0$, the holomorphic sections of $\mathcal{O}(m)$ have northern coefficients equal to the polynomials of degree at most m. Their dimension is $m+1$. For $m < 0$ there is no nonzero holomorphic section.*

Proof. The northern coefficient is an entire function, with Taylor expansion $f_N(z) = \sum_{j\ge 0} a_j z^j$ convergent for every $z \in \mathbb{C}$. The coefficient relation gives the Laurent expansion $f_S(w) = \sum_{j\ge 0} a_j w^{m-j}$ on a punctured neighbourhood of $w = 0$. Recall the elementary uniqueness of a convergent Laurent expansion: its coefficients are fixed by circle integrals, so a holomorphic extension has no negative powers. Thus $a_j = 0$ whenever $j > m$. For $m < 0$ this kills every coefficient. For $m \ge 0$ it leaves exactly $1, z, \dots, z^m$. Conversely each of these monomials gives a nonnegative southern power and hence a global section. Their distinct Taylor coefficients prove linear independence, so the dimension is $m+1$. ∎

To count the other chirality we use *Serre duality*, a cited theorem for compact complex curves, whose sphere case is $H^1(E) \cong H^0(K_{S^2} \otimes E^*)^*$. Here E^* is the dual line and the outer star denotes the dual vector space. See Walpuski, Remark 70.5, p. 77, or Gunning, *Function Theory on Compact Riemann Surfaces*, Theorem 1.17, p. 28. For $E = \mathcal{O}(m)$ this gives $\dim H^1(\mathcal{O}(m)) = \dim H^0(\mathcal{O}(-m-2))$. The polynomial result therefore determines both counts.

Theorem 28.3 (Monopole zero modes · *sketched*)**.** *On the round sphere with the monopole connection on $L_n = \mathcal{O}(n)$, for every $n \in \mathbb{Z}$,*

$$\dim\ker D^+ = \max(n,0), \qquad \dim\ker D^- = \max(-n,0), \qquad \operatorname{ind} D^+ = n.$$

The total number of zero modes is $|n|$. For $n > 0$ they all have positive chirality. For $n < 0$ they all have negative chirality. For $n = 0$ both kernels vanish for this sphere operator.

Proof. The positive component is $E_n = \mathcal{O}(n-1)$, so the polynomial count gives $\max(n,0)$. Serre duality identifies the negative count with $\dim H^0(\mathcal{O}(-n-1)) = \max(-n,0)$. Subtracting gives n for every integer n. Adding gives $|n|$. The global kernel identifications are the cited analytic steps in this derivation. ∎

Example 28.4 (Three zero modes in a monopole field)**.** Take the sphere monopole with $n = 3$ and the nonnegative Pauli operator $H = D^2$. An overall positive physical energy scale would not change its zero modes. Find its zero-energy states and their chirality. For a smooth spinor Ψ, the identity $\langle \Psi, D^2\Psi\rangle = \|D\Psi\|^2$ shows that $\ker H = \ker D$. The theorem gives three positive modes and no negative mode. Here the positive line has $m = n - 1 = 2$.

For a general $m \geq 0$, use Lecture 22's unit frame for $O(m)$. The local positive zero-mode equation and its monomial solutions are

$$\left(\partial_{\bar{z}} + \frac{mz}{2(1+|z|^2)}\right)\psi = 0, \qquad \psi_j(z) = (1+|z|^2)^{-m/2} z^j, \quad 0 \leq j \leq m.$$

Differentiating the weight in ψ_j gives $-mz\psi_j/[2(1+|z|^2)]$, which cancels the connection term. These are unit-frame coefficients of sections of E_n. They are not coefficients relative to $\mathrm{d}z^{1/2}$ or to a coordinate density. The spin connection has already entered through the degree shift from n to m. For $m = 2$ the northern coefficients are

$$\frac{1}{1+|z|^2}, \qquad \frac{z}{1+|z|^2}, \qquad \frac{z^2}{1+|z|^2}.$$

Their southern holomorphic coefficients are w^2, w and 1, respectively, so each section is holomorphic at the south pole.

For completeness, their squared norms use the round area $\mathrm{d}A = 4\mathrm{d}x\mathrm{d}y/(1+|z|^2)^2$, with $z = x + \mathrm{i}y$. The angular integral makes distinct monomials orthogonal. With $r = |z|$, the three radial norms are

$$8\pi \int_0^\infty \frac{r^{2j+1}}{(1+r^2)^4}\mathrm{d}r = \begin{cases} 4\pi/3, & j = 0, 2, \\ 2\pi/3, & j = 1. \end{cases}$$

Substitution $t = r^2$ reduces these to the integrals of $4\pi t^j/(1+t)^4$. Next put $u = (1+t)^{-1}$. The three integrals divided by 4π become $\int_0^1 u^2\mathrm{d}u = 1/3$, $\int_0^1 u(1-u)\mathrm{d}u = 1/6$ and $\int_0^1 (1-u)^2\mathrm{d}u = 1/3$, giving the displayed values. Dividing each coefficient by the square root of its stated squared norm gives an orthonormal basis. All three are zero-energy states of positive chirality.

The *lowest Landau level* is the lowest scalar magnetic-kinetic eigenspace. Haldane's spinless sphere model has $n+1$ lowest-level states for a nonnegative monopole magnitude n. A consulted derivation is Greiter, equations (2) and (20)–(21). His magnitude is $2s_0$, and the lowest level has angular momentum s_0, hence $2s_0 + 1$ states. His electron charge is negative. Here this comparison uses the flux magnitude, not a borrowed signed Chern convention. Its holomorphic description uses $O(n)$, rather than the spinor line $O(n-1)$. The corresponding polynomial count is $n+1$, giving four states when $n = 3$. The spin connection accounts for this one-state difference. This comparison concerns degeneracy. The scalar lowest Landau energy is not being set equal to the Dirac zero energy.

28.4 The surface index theorem

The polynomial count determines the left side of an index theorem for this sphere connection. The right side is the flux integer already computed in Lecture 22.

Theorem 28.5 (Atiyah–Singer on a surface · *cited*). *Let M be a closed oriented Riemannian spin surface and L a Hermitian line bundle with unitary connection of curvature F_{conn}. Choose positive spinors in the square root of the canonical line for the chosen spin structure. For the corresponding chiral twisted Dirac operator,*

$$\operatorname{ind} D^+ = \int_M c_1(L) = \frac{\mathrm{i}}{2\pi}\int_M F_{\mathrm{conn}}.$$

This is the surface specialization of the twisted Dirac index theorem (Nakahara, equations (12.85)–(12.86)). The curvature here is anti-Hermitian, as in Lecture 22. For the sphere monopole, $F_{\mathrm{conn}} = -\mathrm{i}n \sin\theta \, \mathrm{d}\theta \wedge \mathrm{d}\varphi/2$. Its curvature integral gives $\mathrm{i}(-2\pi\mathrm{i}n)/(2\pi) = n$, agreeing with the kernel count. The theorem applies beyond this symmetric monopole connection. It fixes the difference of the kernels under smooth changes of the metric and connection on the same graded bundle. It does not generally fix either kernel dimension separately. An arbitrary prescribed pair of nonnegative dimensions need not be realizable by such a family.

28.5 The two-dimensional chiral anomaly

Remark 28.6 (Axial charge and index · *cited*)**.** For this application reinterpret two dimensions as Euclidean spacetime. The spatial-first coordinates are (x, τ) with orientation $\mathrm{d}x \wedge \mathrm{d}\tau$ and $z = x + \mathrm{i}\tau$. The Wick rotation is $t = -\mathrm{i}\tau$, so the local holomorphic argument becomes $z = x - t$. A fixed value of $x - t$ moves toward increasing x. Positive Euclidean chirality therefore continues to right movers. This local convention matches the Euclidean grading to the Lorentzian one.

For one massless Dirac fermion let j_R and j_L be the right and left number currents. Define the axial current by $j_5 = j_R - j_L$. Let Q_5 be its spatially integrated charge. With the vector current conserved, the cited integrated anomaly relation is

$$\Delta Q_5 = 2 \operatorname{ind} D^+.$$

It concerns the index of the corresponding Euclidean gauge background and assumes no axial charge escapes through the spatial boundary during the transition. It is not inferred by treating the spatial monopole sphere as Lorentzian spacetime. A background of index +1 gives a change +2 in this axial convention. The factor two comes from the Dirac fermion's two chiral contributions to the axial imbalance. Bilal prints the current relation in four dimensions: (3.30) in §3.5, p. 14, has divergence $-a$, and (3.41) in §3.7, p. 16, has integrated anomaly function -2 times the index, provided the Euclidean and Minkowski chiralities are matched. The regulated chirality-trace argument (3.39)–(3.40) is independent of dimension. Nonzero modes cancel in pairs. The zero modes contribute their signed difference. The fermion and antifermion measures supply the factor two. Repeating that argument with the two-dimensional measure gives the stated relation. The two minus signs give its positive factor. Bilal extends the index computation to arbitrary even dimension in (3.44)–(3.47). This is the cited quantum-field-theory input for the two-dimensional remark, rather than a claim that his printed (3.41) is a two-dimensional formula.

Self-check. For $n = 0$, determine the sphere index and both zero-mode counts. Explain what follows from the index alone.

Answer. The sphere theorem gives index zero and both kernel dimensions zero. The index alone says only that the two dimensions agree. Their vanishing uses the sphere-specific polynomial and Serre-duality count.

Summary. The positive spinor line on the outward sphere is $\mathcal{O}(-1)$. Twisting by the monopole line $\mathcal{O}(n)$ gives $\mathcal{O}(n-1)$. The polynomial proof and cited spinor, Hodge and Serre identifications give chiral counts $\max(n, 0)$ and $\max(-n, 0)$. Their difference is n, agreeing with the cited surface index theorem and the flux Chern number. For positive n, the spinor

count is n and the spinless lowest Landau-level count is $n+1$. With the stated spacetime, chirality and boundary conventions, the cited axial anomaly gives $\Delta Q_5 = 2\,\mathrm{ind}\,D^+$. The course's topological integers thus constrain both classical boundary sectors and quantum zero modes.

Reading. Lawson–Michelsohn, Chapter III, §5, pp. 192–197, and §7, pp. 201–202; Chapter IV, §11; Appendix D, p. 396. Walpuski, *Differential Geometry IV* (2022), Propositions 70.3 and 70.6 and Remark 70.5, pp. 76–78. Gunning, *Function Theory on Compact Riemann Surfaces*, Theorem 1.17, p. 28. Nakahara, §12.6. Bilal, §3.5–3.7, pp. 13–17. Haldane, *Phys. Rev. Lett.* 51 (1983), 605–608; Greiter, *Phys. Rev. B* 83 (2011), 115129, §II.